



Operating Systems

Lecturer

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Scheduling Criteria

- **CPU utilization** – keep the CPU as busy as possible
- **Throughput** – # of processes that complete their execution per time unit
- **Turnaround time** – amount of time to execute a particular process
- **Waiting time** – amount of time a process has been waiting in the ready queue
- **Response time** – amount of time it takes from when a request was submitted until the first response is produced, not output (for time-sharing environment)

Scheduling Algorithm Optimization Criteria

- **Maximize CPU utilization**
- **Maximize throughput**
- **Minimize turnaround time**
- **Minimize waiting time**
- **Minimize response time**

Scheduling Algorithms

There are many different CPU-Scheduling algorithm

1. First Come, First Served (FCFS).
2. Shortest job First (SJF).
 - Preemptive SJF
 - Non-Preemptive SJF
3. Priority
4. Round Robin
5. Multilevel Queues

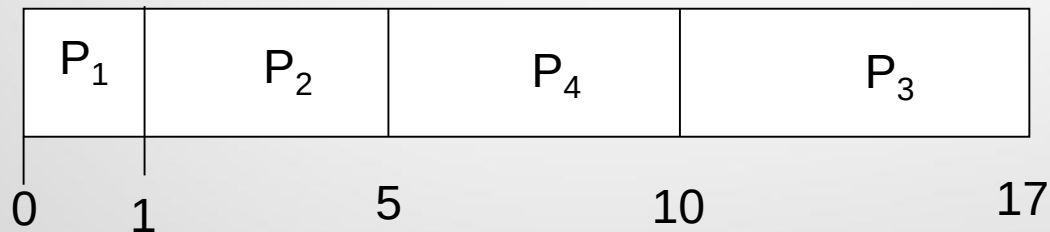
Example of Shortest-Job-first-time SJF

Non-Preemptive SJF

- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	1
P_2	1	4
P_3	2	7
P_4	3	5

- Non-Preemptive SJF Gantt Chart*



- Average waiting time = $[(0-0)+(1-1)+(10-2)+5-3)]/4 = 26/4 = 2.5$ msec

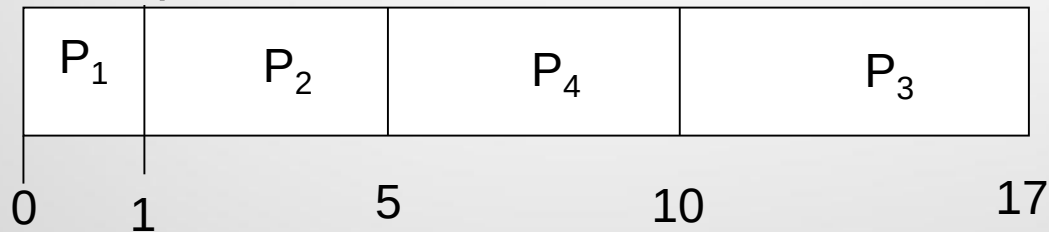
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Non-Preemptive SJF

- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	1
P_2	1	4
P_3	2	7
P_4	3	5

- Non-Preemptive SJF Gantt Chart*



Waiting time

P_1	P_2	P_3	P_4
0-0 = 0	1-1 = 0	10-2 = 8	5-3 = 2

Turnaround time

P_1	P_2	P_3	P_4
0-1= 1	1-5= 4	17-2= 15	10-3= 7

Response time

P_1	P_2	P_3	P_4
0	0	10-2= 8	5-3= 2

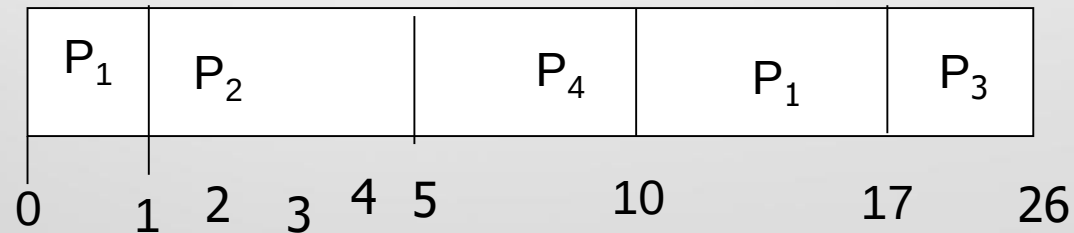
Example of Shortest-Job-first-time SJF

Preemptive SJF

- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

- Preemptive SJF Gantt Chart*



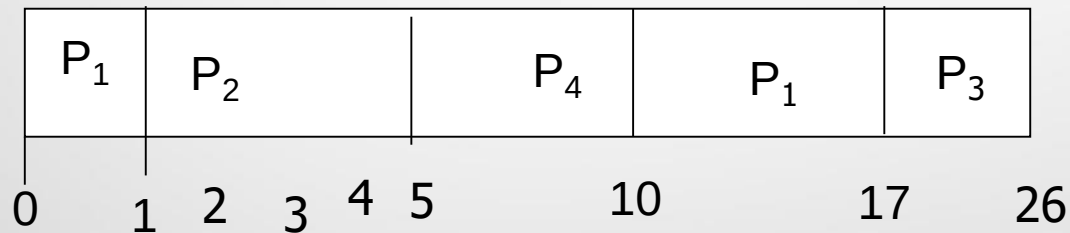
Example of Shortest-Job-first-time SJF

Preemptive SJF

- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

- Preemptive SJF** Gantt Chart



Waiting time			
P_1	P_2	P_3	P_4
10-1=9	1-1=0	17-2=15	5-3=2

Turnaround time			
P_1	P_2	P_3	P_4
17-0=17	5-1=4	26-2=24	10-3=7

Response time			
P_1	P_2	P_3	P_4
0-0=0	1-1=0	17-2=15	5-3=2

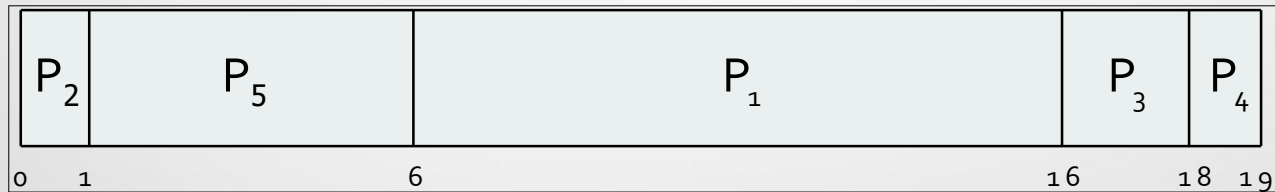
Priority Scheduling

- A priority number (integer) is associated with each process
- The CPU is allocated to the process with the highest priority (smallest integer $\equiv$ highest priority)
 - Preemptive
 - Nonpreemptive
- SJF is priority scheduling where priority is the inverse of predicted next CPU burst time
- Problem $\equiv$ Starvation – low priority processes may never execute
- Solution $\equiv$ Aging – as time progresses increase the priority of the process

Example of Priority Scheduling

<u>Process</u>	<u>Burst Time</u>	<u>Priority</u>
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2

- Priority scheduling Gantt Chart



- Average waiting time = 8.2 msec

Waiting time				
P_1	P_2	P_3	P_4	P_5
-6 =0 6	0- 0= 0	16- 0= 16	18- 0= 18	1- 0= 1

Turnaround Time				
P_1	P_2	P_3	P_4	P_5
16	1	18	19	6

Respond time				
P_1	P_2	P_3	P_4	P_5
0-6 = 6	0- 0= 0	16- 0= 16	18- 0= 18	1- 0= 1

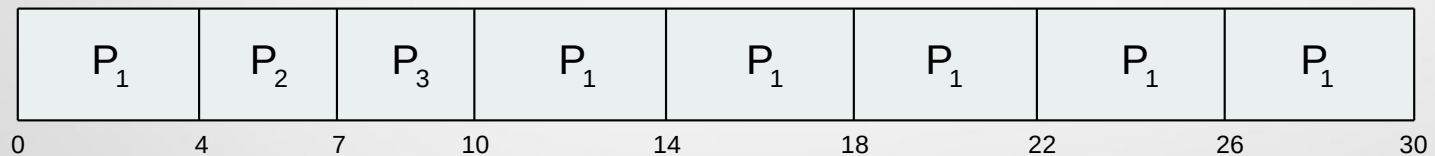
Round Robin (RR)

- Each process gets a small unit of CPU time (**time quantum q**), usually 10-100 milliseconds. After this time has elapsed, the process is preempted and added to the end of the ready queue.
- If there are n processes in the ready queue and the time quantum is q , then each process gets $1/n$ of the CPU time in chunks of at most q time units at once.
- Max. No process waits more than $(n-1)q$ time units.
- Timer interrupts every quantum to schedule next process
- Performance
 - q large $\Rightarrow$ FIFO
 - q small $\Rightarrow q$ must be large with respect to context switch, otherwise overhead is too high

Example of RR with Time Quantum = 4

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- All the processes arrive at the same time 0.
- Round Robin (RR) Scheduling of quantum: 4 ms.
- The Gantt chart is:

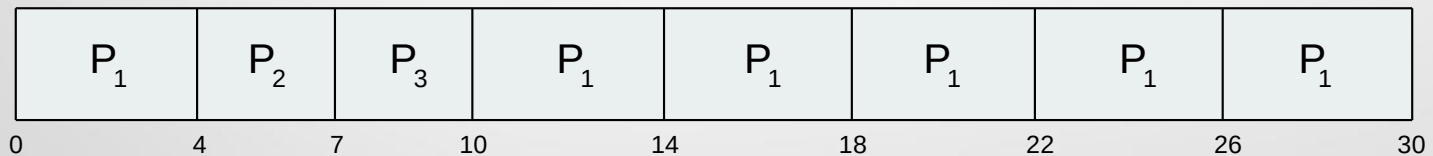


- of context switches = ???

Example of RR with Time Quantum = 4

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- All the processes arrive at the same time 0.
- Round Robin (RR) Scheduling of quantum: 4 ms.
- The Gantt chart is:



- of context switches = 7

Example of RR with Time Quantum = 4

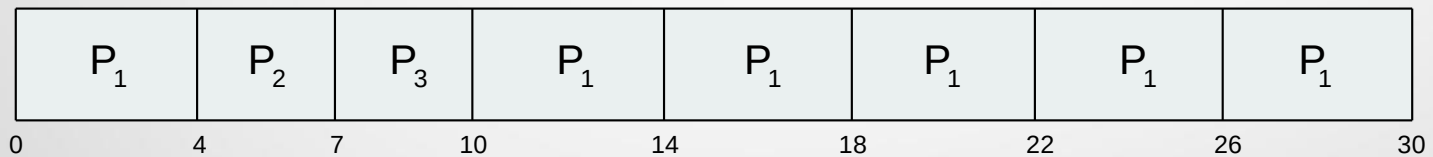
<u>Process</u>	<u>Burst Time</u>
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P_1	24
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P_2	3
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P_3	3
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- All the processes arrive at the same time 0.
- Round Robin (RR) Scheduling of quantum: 4 ms.
- The Gantt chart is:



Waiting time		
P_1	P_2	P_3
10-4=6	4	7

Turnaround time		
P_1	P_2	P_3
30-0=30	7-0 = 7	10-0=10

Response time		
P_1	P_2	P_3
0-0=0	4-0=4	7-0=7

Avg. Waiting = 5.667 ms

Avg. Turnaround = 15.667 ms

Avg. Respond = 3.667 ms

Multilevel Queue

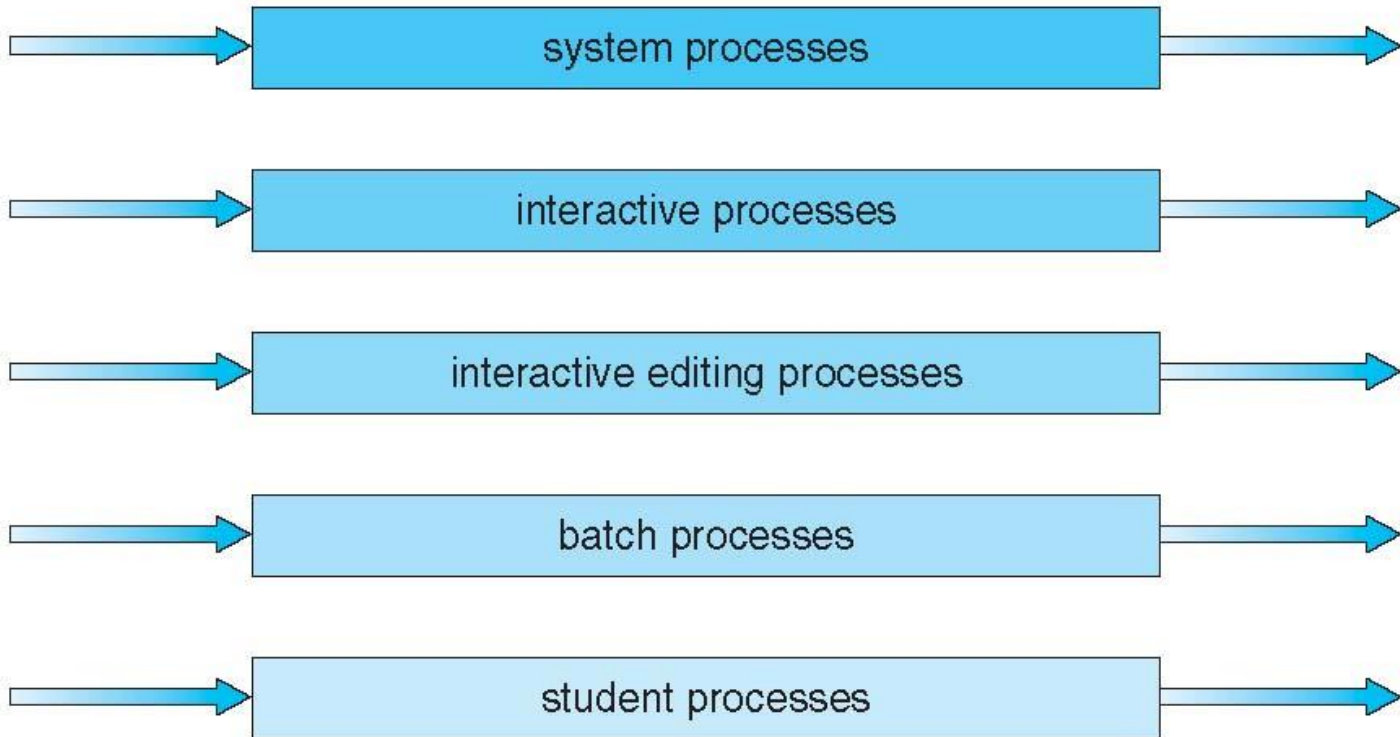
- Ready queue is partitioned into separate queues, eg:
 - **foreground** (interactive)
 - **background** (batch)
- Process permanently in a given queue
- Each queue has its own scheduling algorithm:
 - foreground – RR
 - background – FCFS

Multilevel Queue

- Scheduling must be done between the queues:
 - Fixed priority scheduling; (i.e., serve all from foreground then from background). Possibility of starvation.
 - Time slice – each queue gets a certain amount of CPU time which it can schedule amongst its processes; i.e., 80% to foreground in RR
 - 20% to background in FCFS

Multilevel Queue Scheduling

highest priority



lowest priority

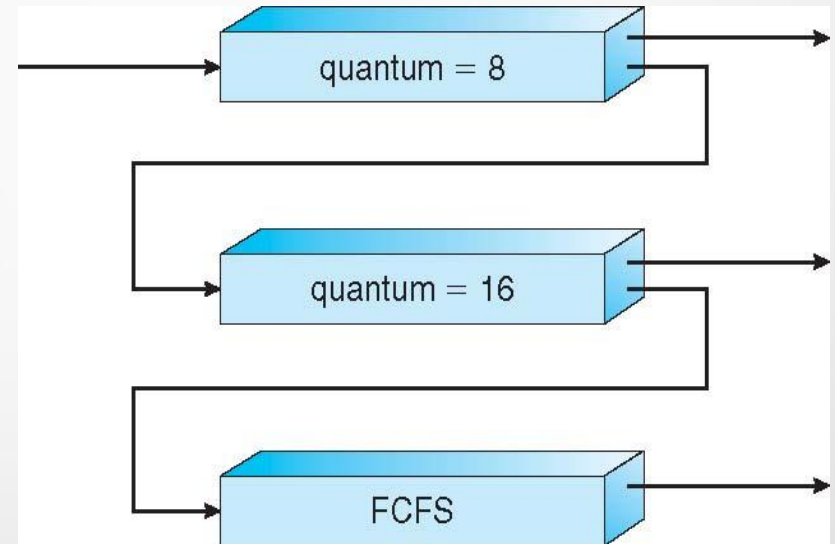
Example of Multilevel Feedback Queue

- **Three queues:**

- Q_0 – RR with time quantum 8 milliseconds
- Q_1 – RR time quantum 16 milliseconds
- Q_2 – FCFS

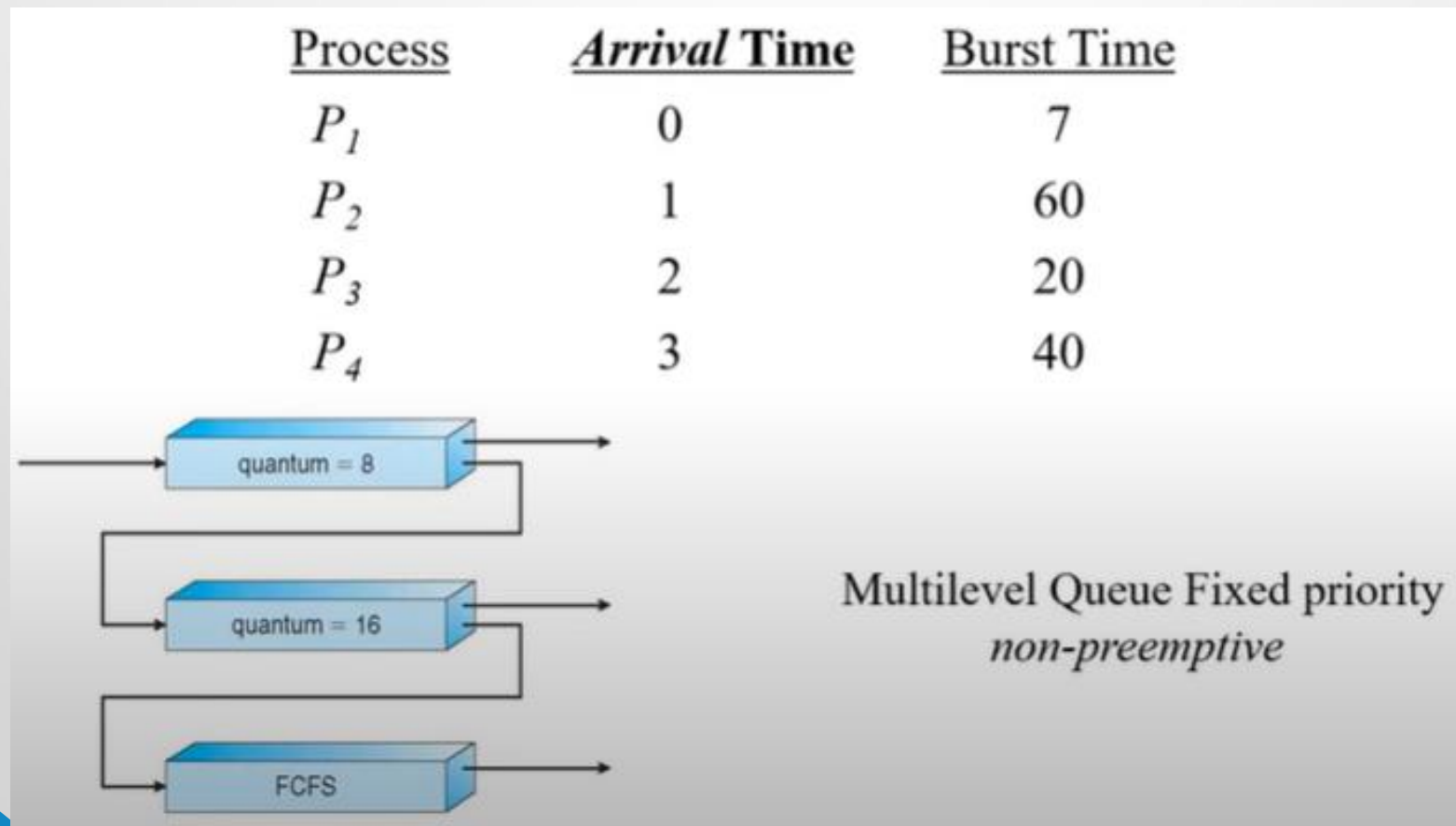
- **Scheduling**

- A new job enters queue Q_0 which is served
 - When it gains CPU, job receives 8 milliseconds
 - If it does not finish in 8 milliseconds, job is moved to queue Q_1
- At Q_1 job is again served FCFS and receives 16 additional milliseconds
 - If it still does not complete, it is preempted and moved to queue Q_2



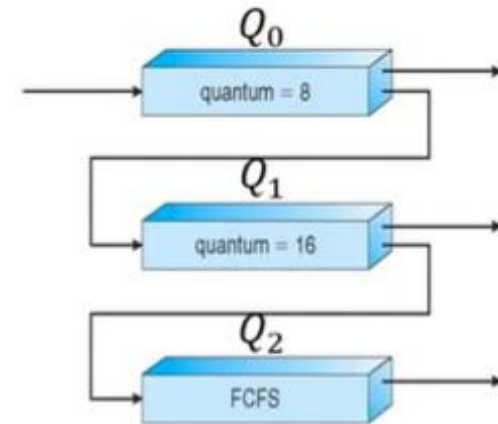
Example of Multilevel Feedback Queue

- Now we add the concepts of varying arrival times and preemption to the analysis



Example of Multilevel Feedback Queue

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	7
P_2	1	60
P_3	2	20
P_4	3	40



Q_0



Q_1

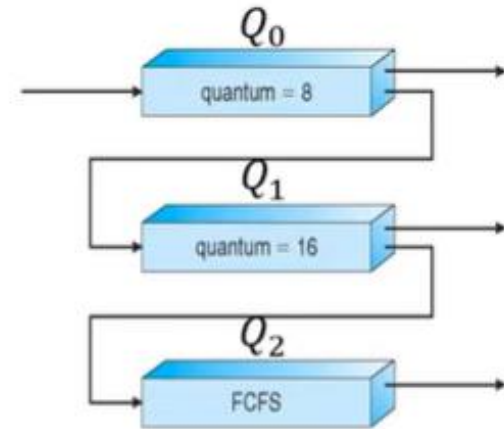


Q_2



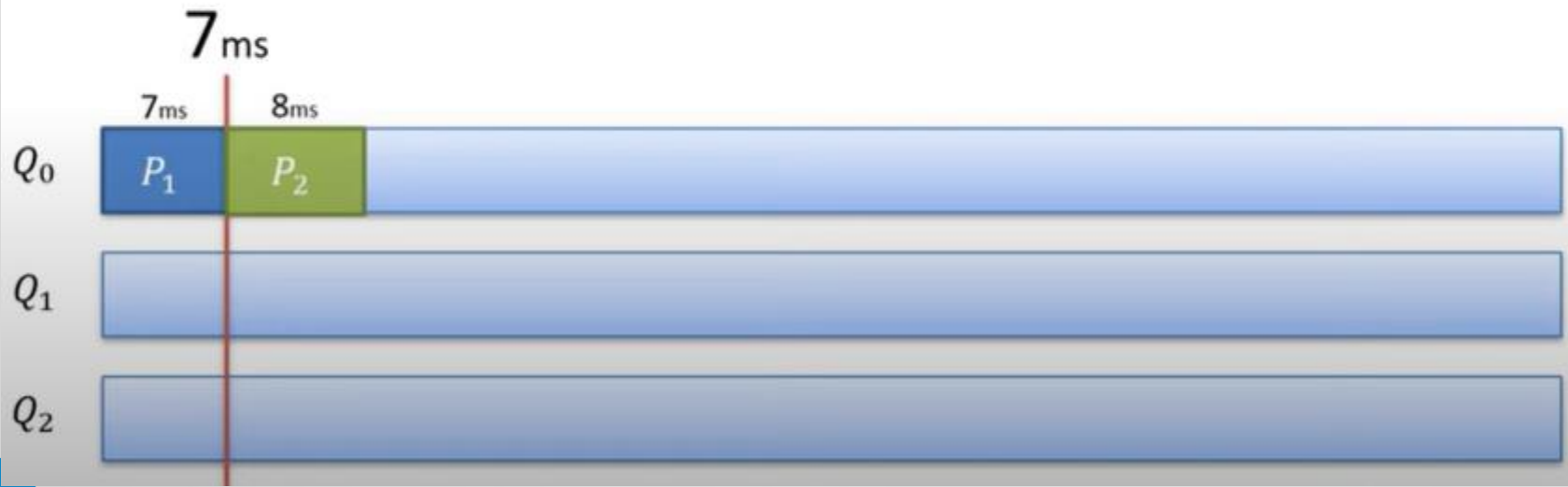
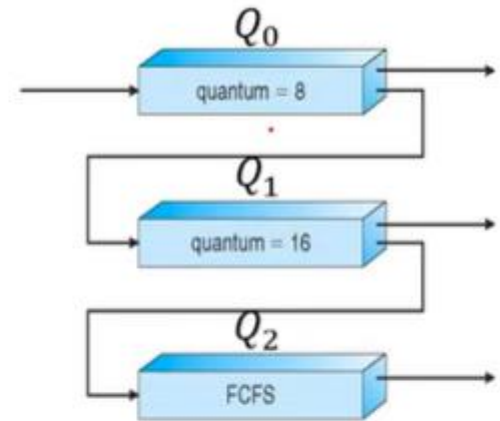
Example of Multilevel Feedback Queue

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	7
P_2	1	60
P_3	2	20
P_4	3	40

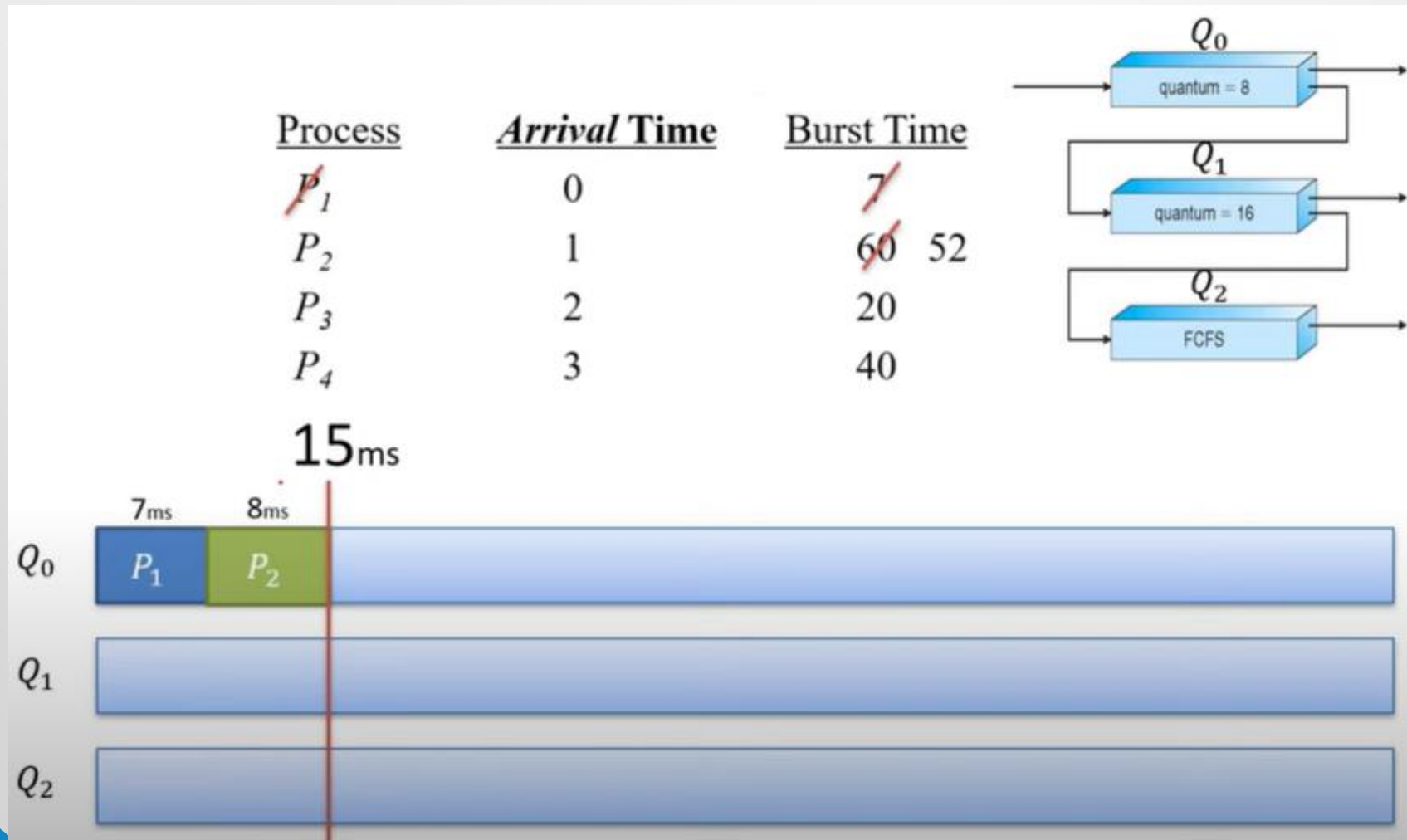


Example of Multilevel Feedback Queue

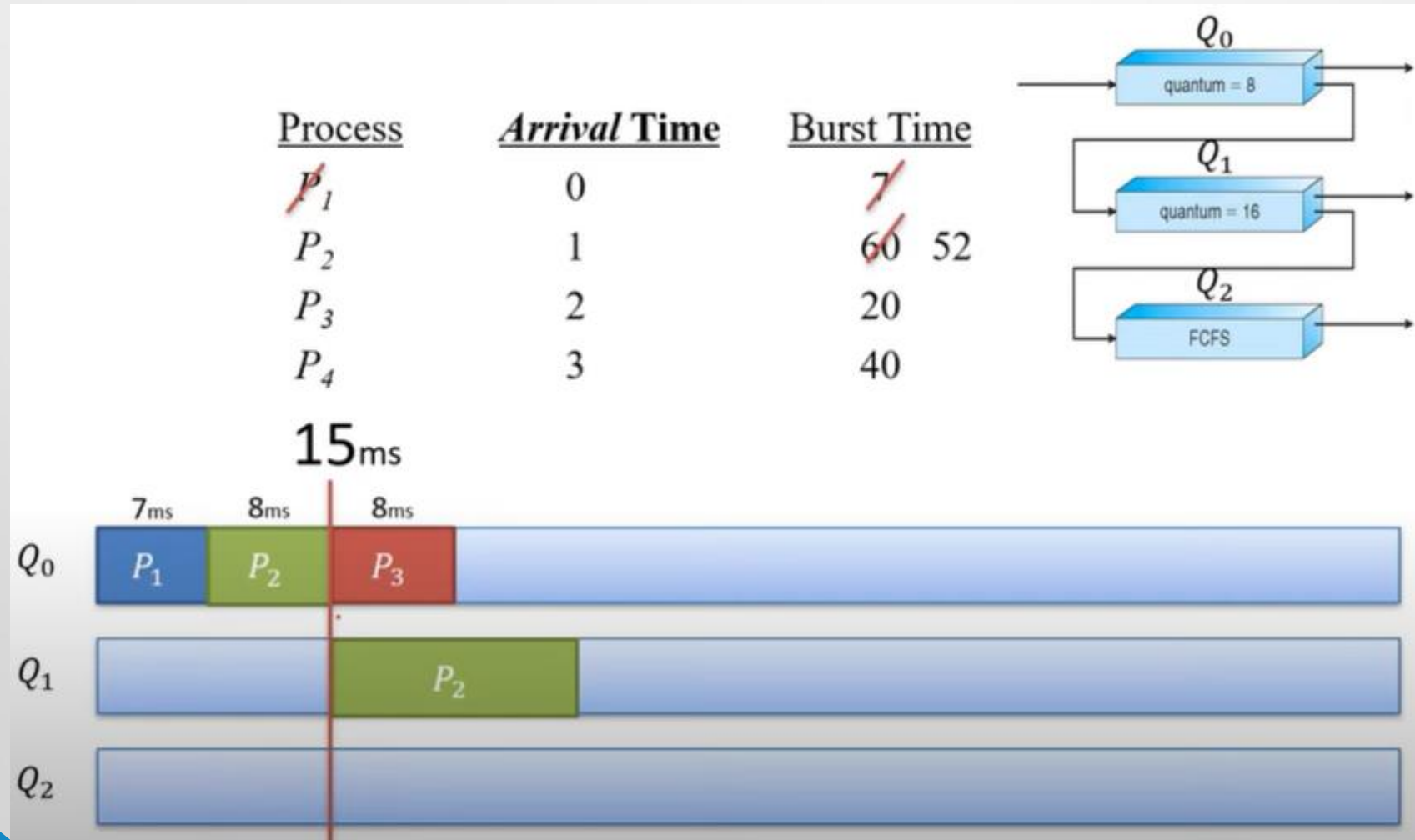
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Example of Multilevel Feedback Queue



Example of Multilevel Feedback Queue



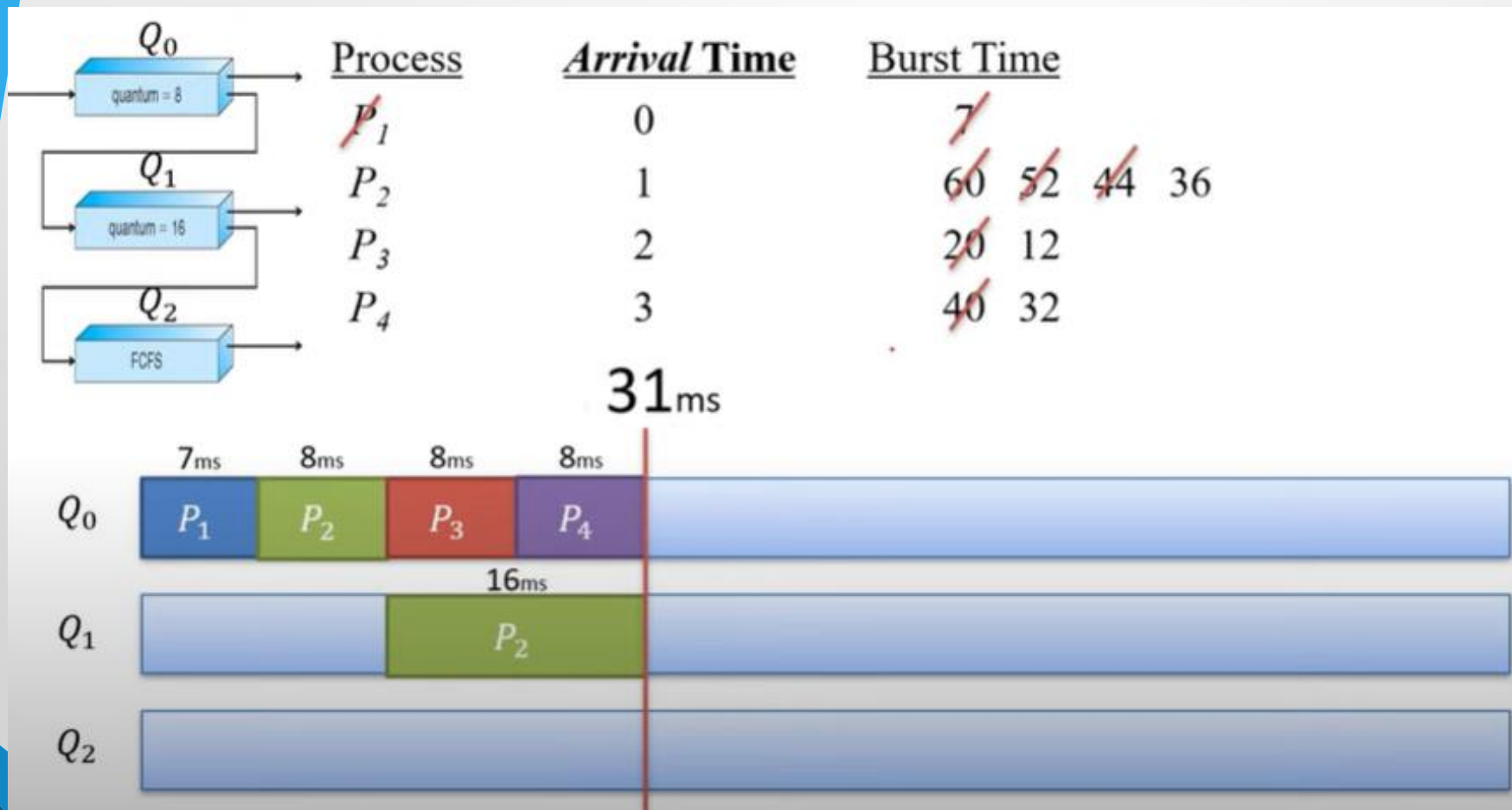
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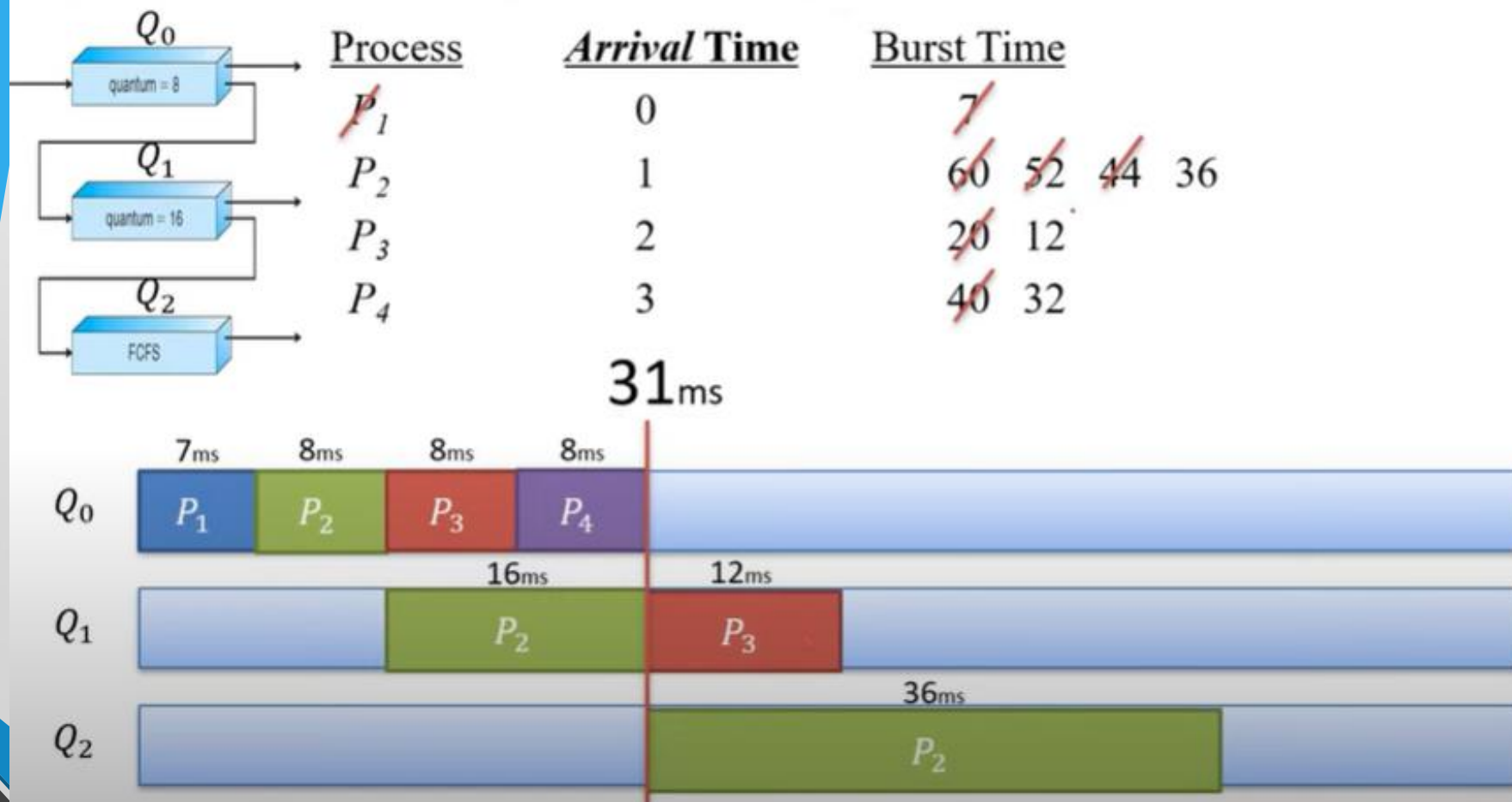
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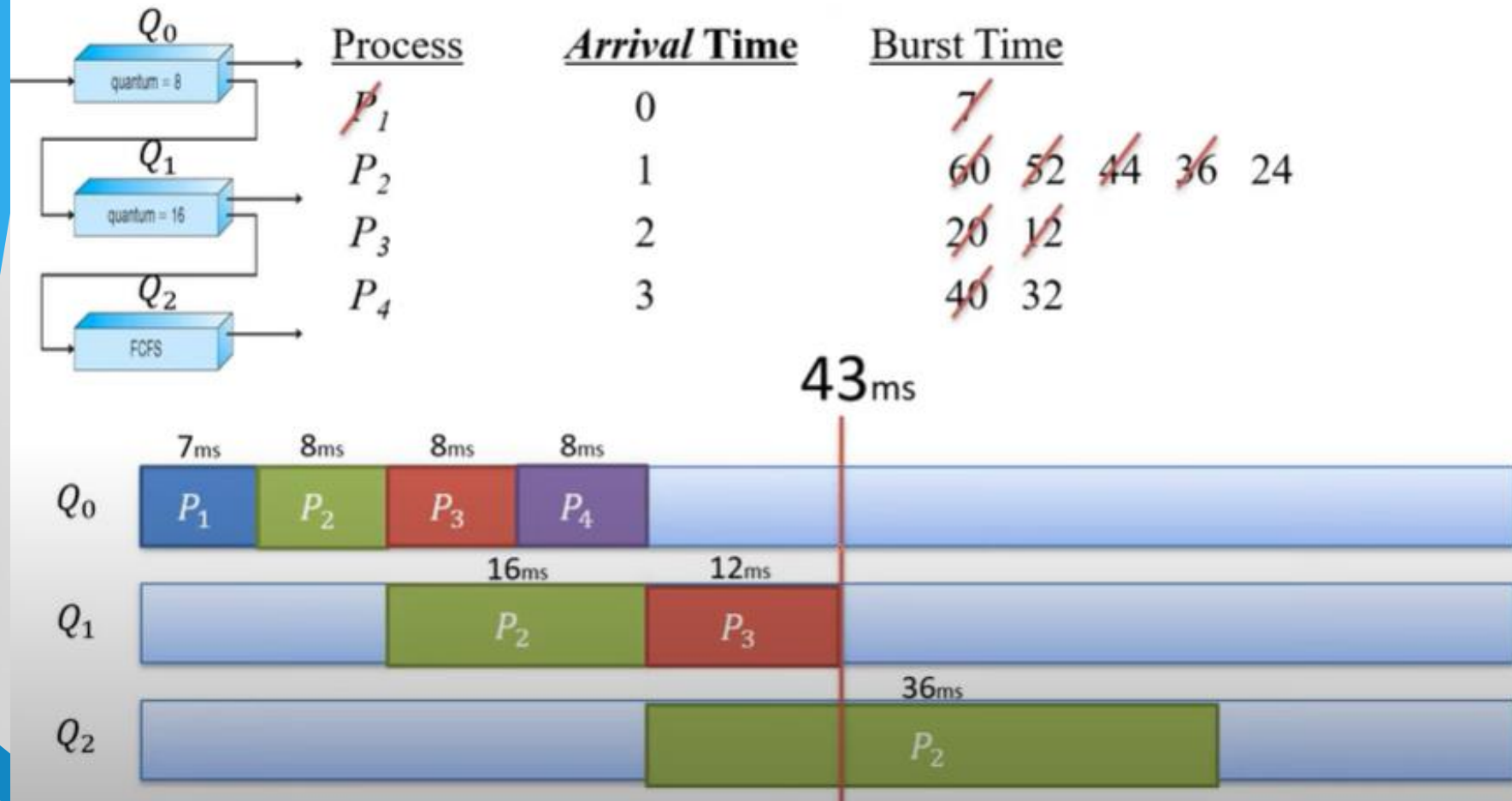
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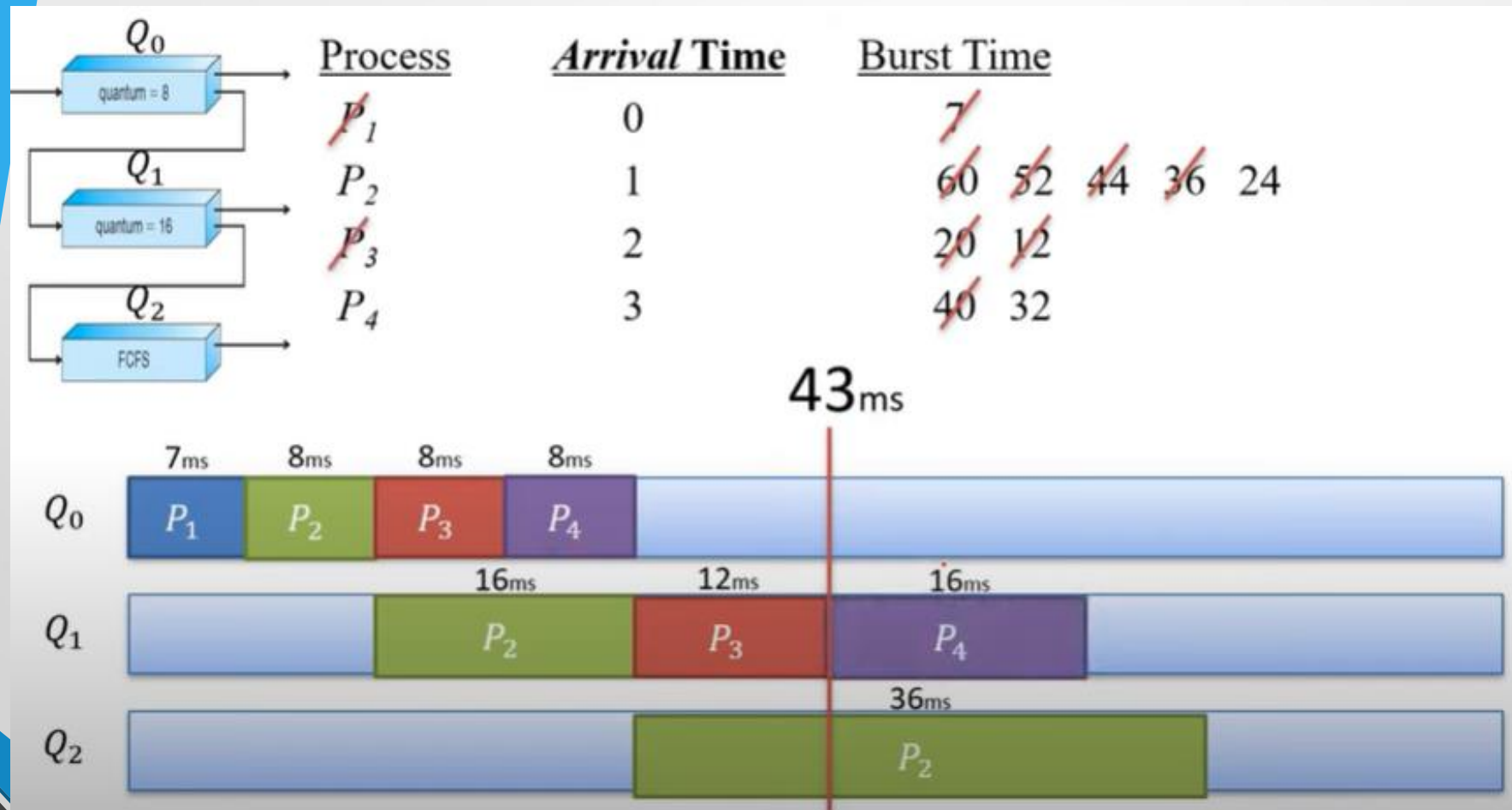
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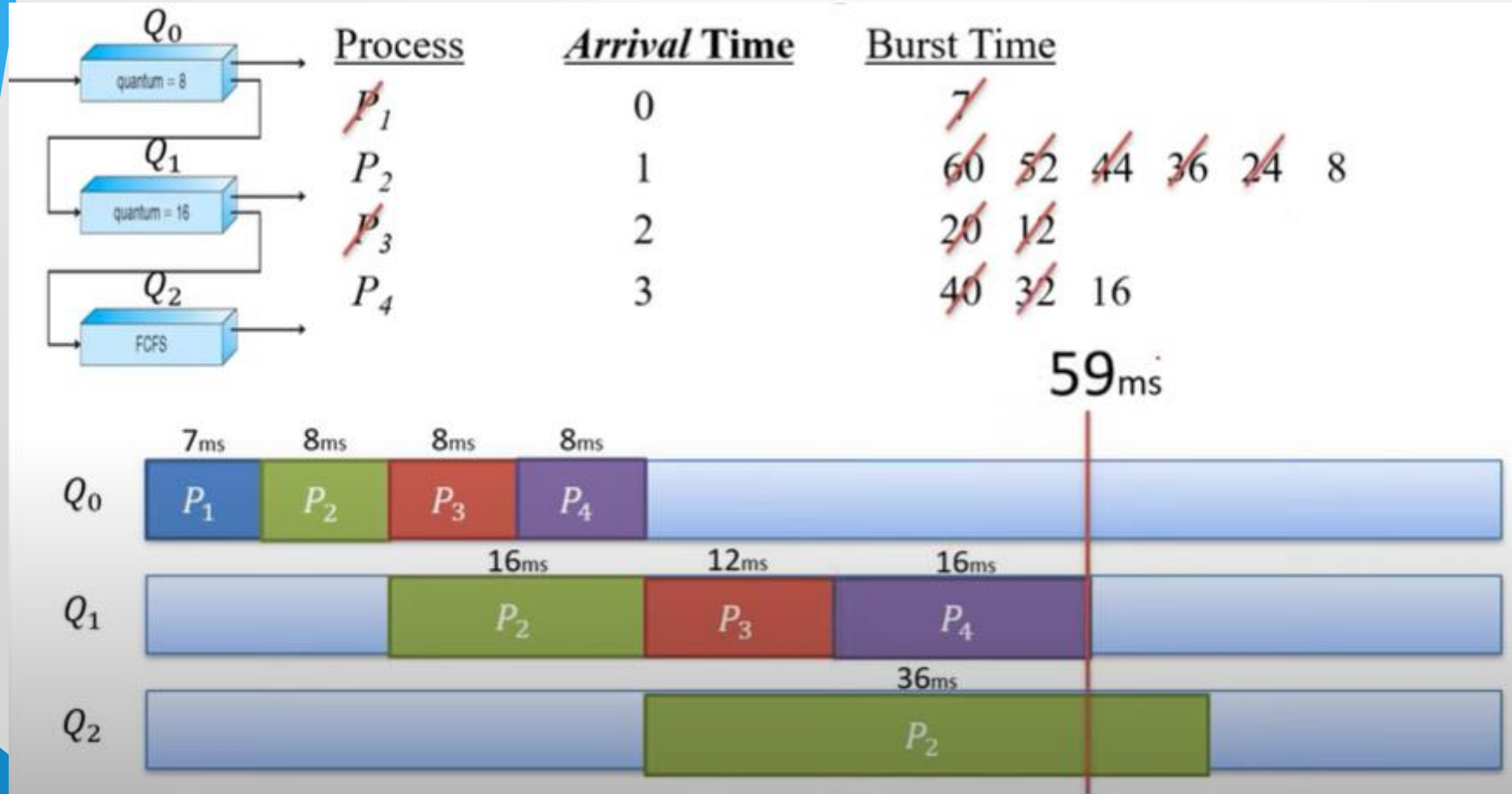
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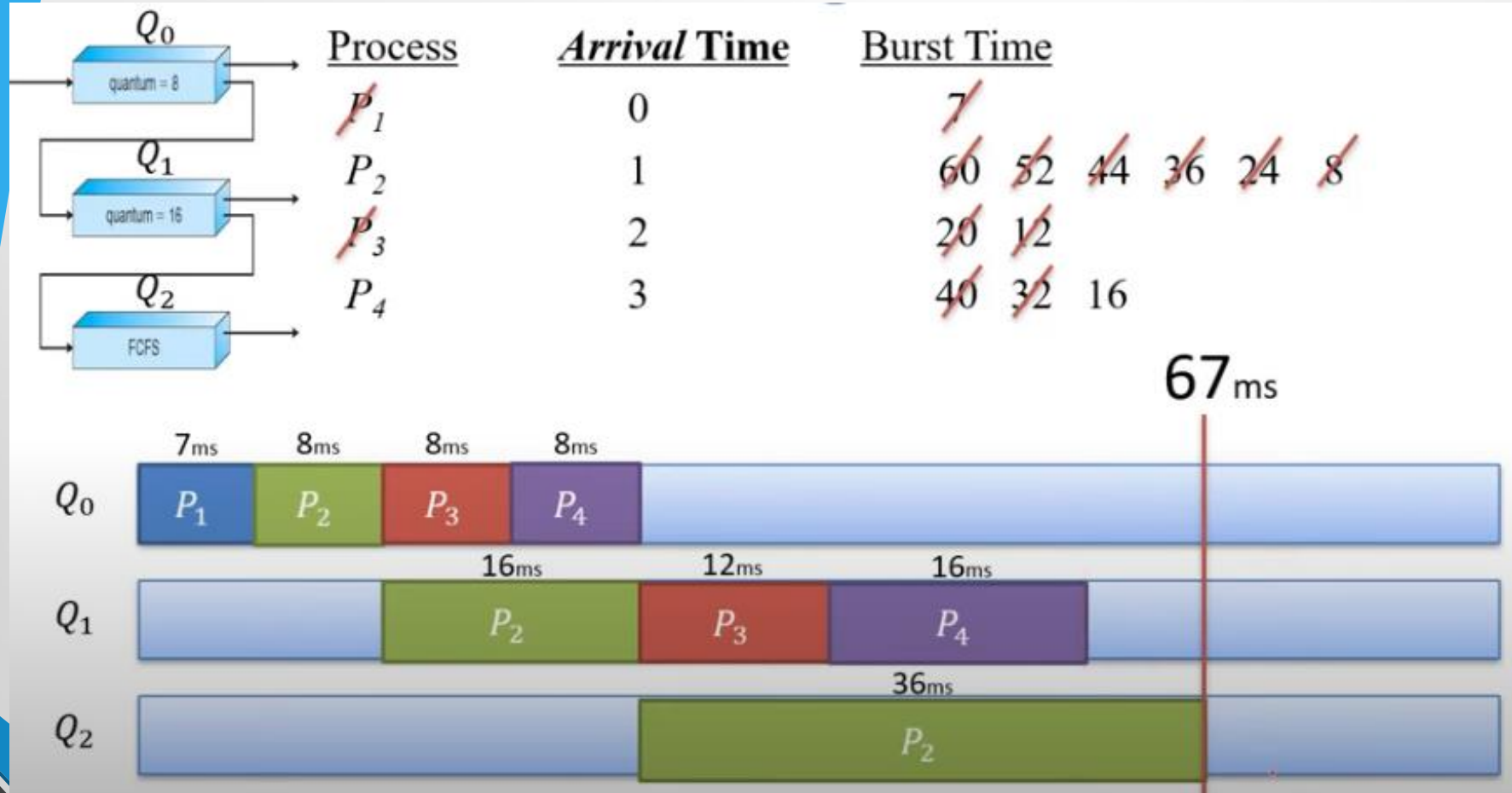
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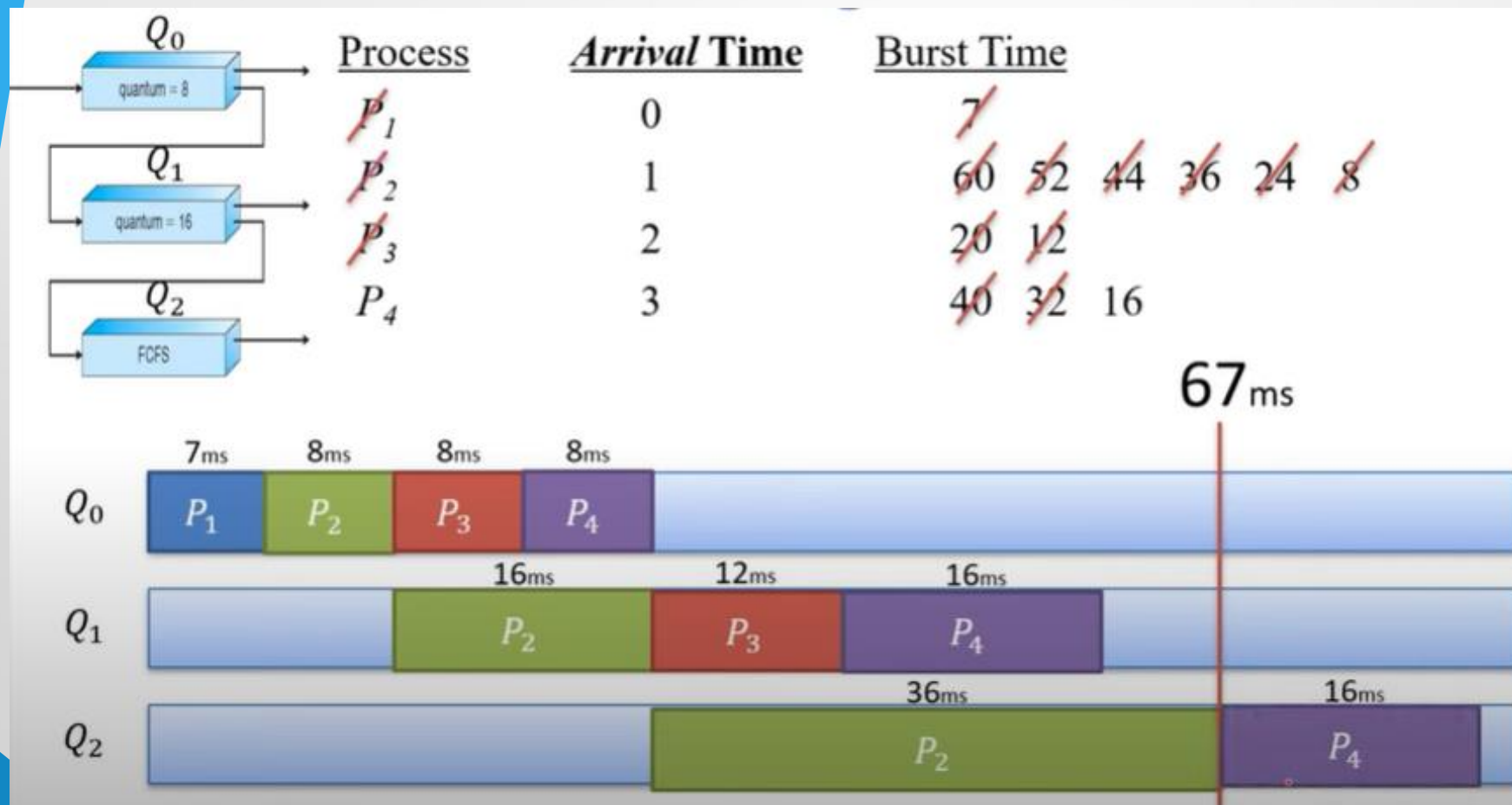
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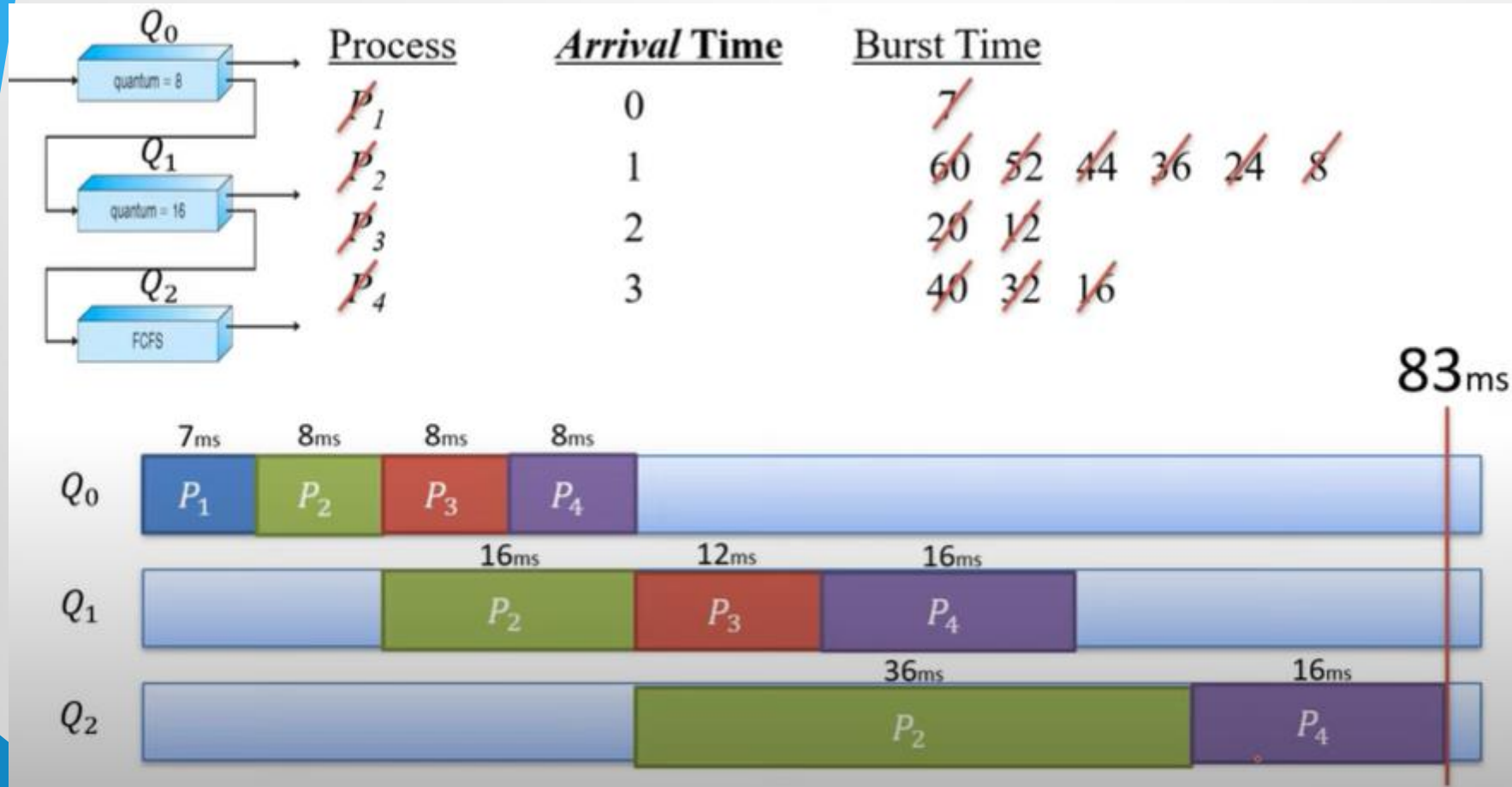
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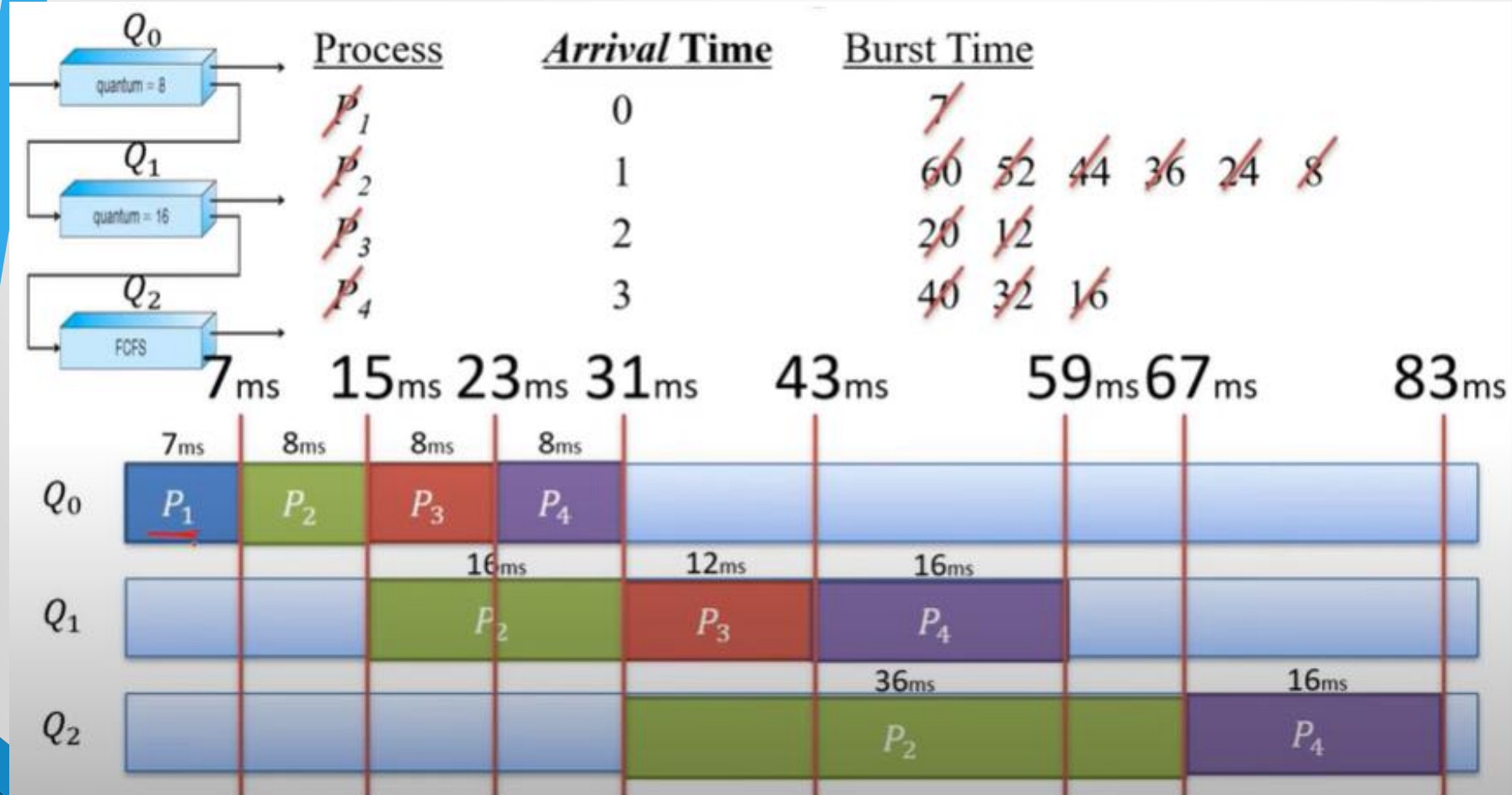
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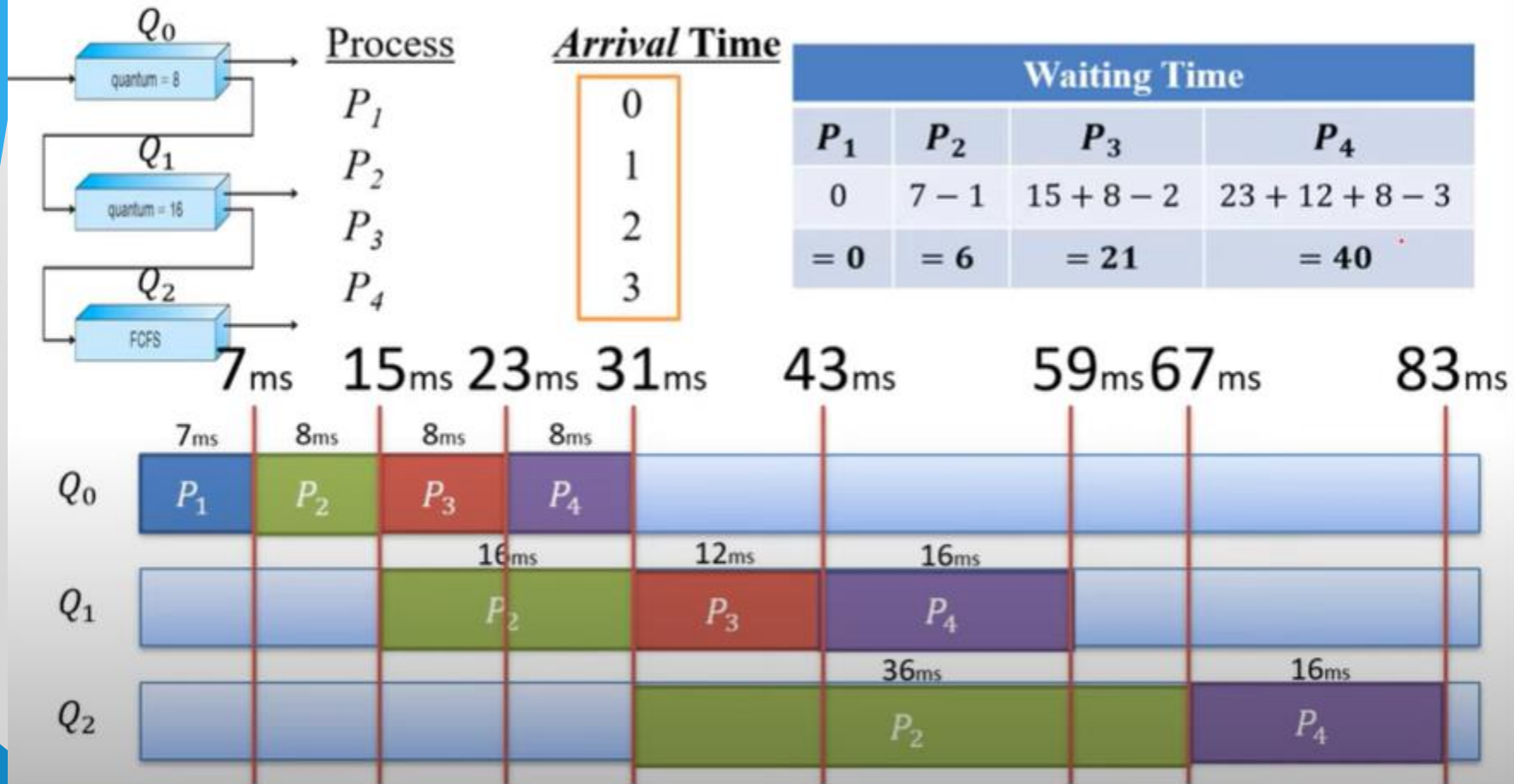
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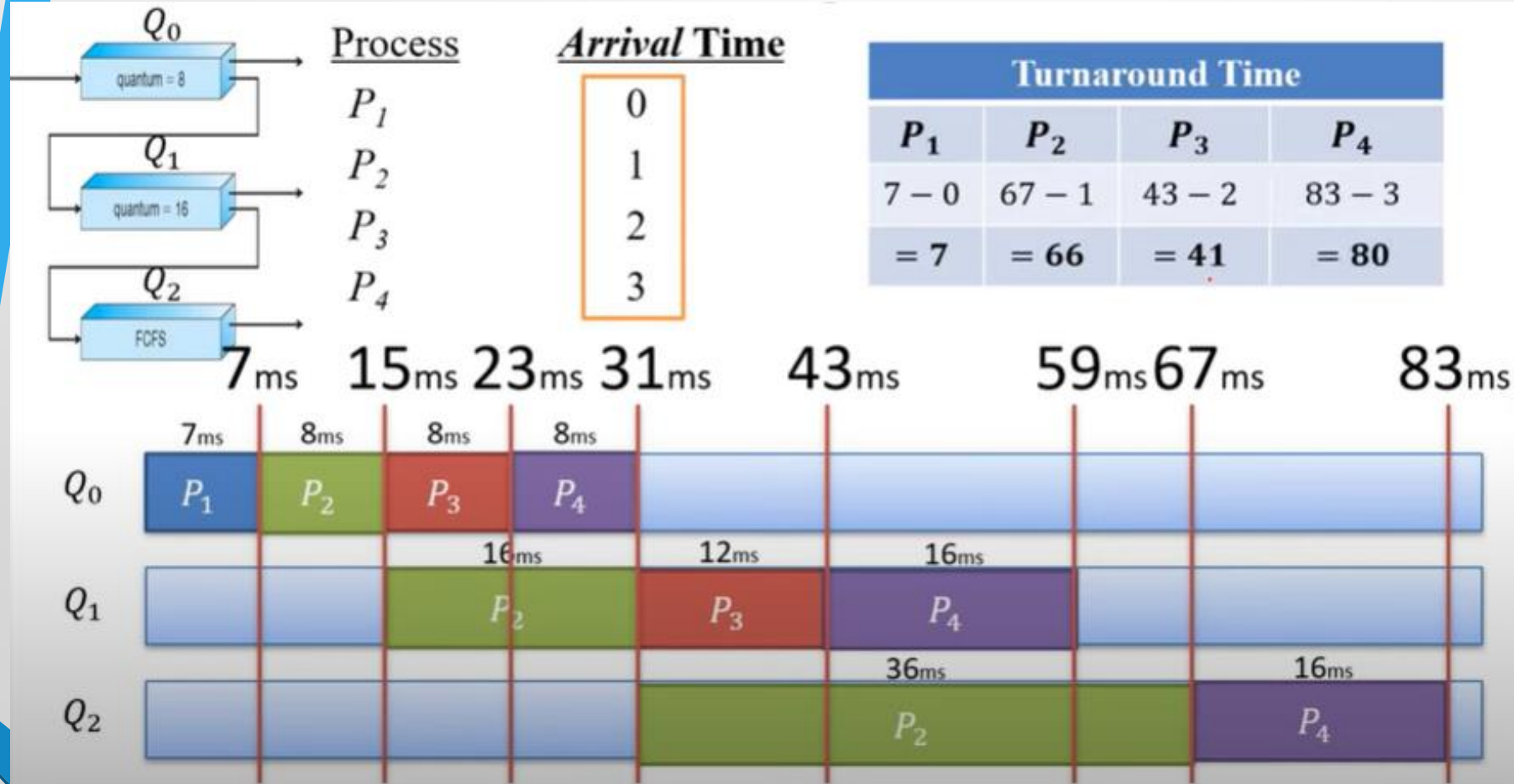
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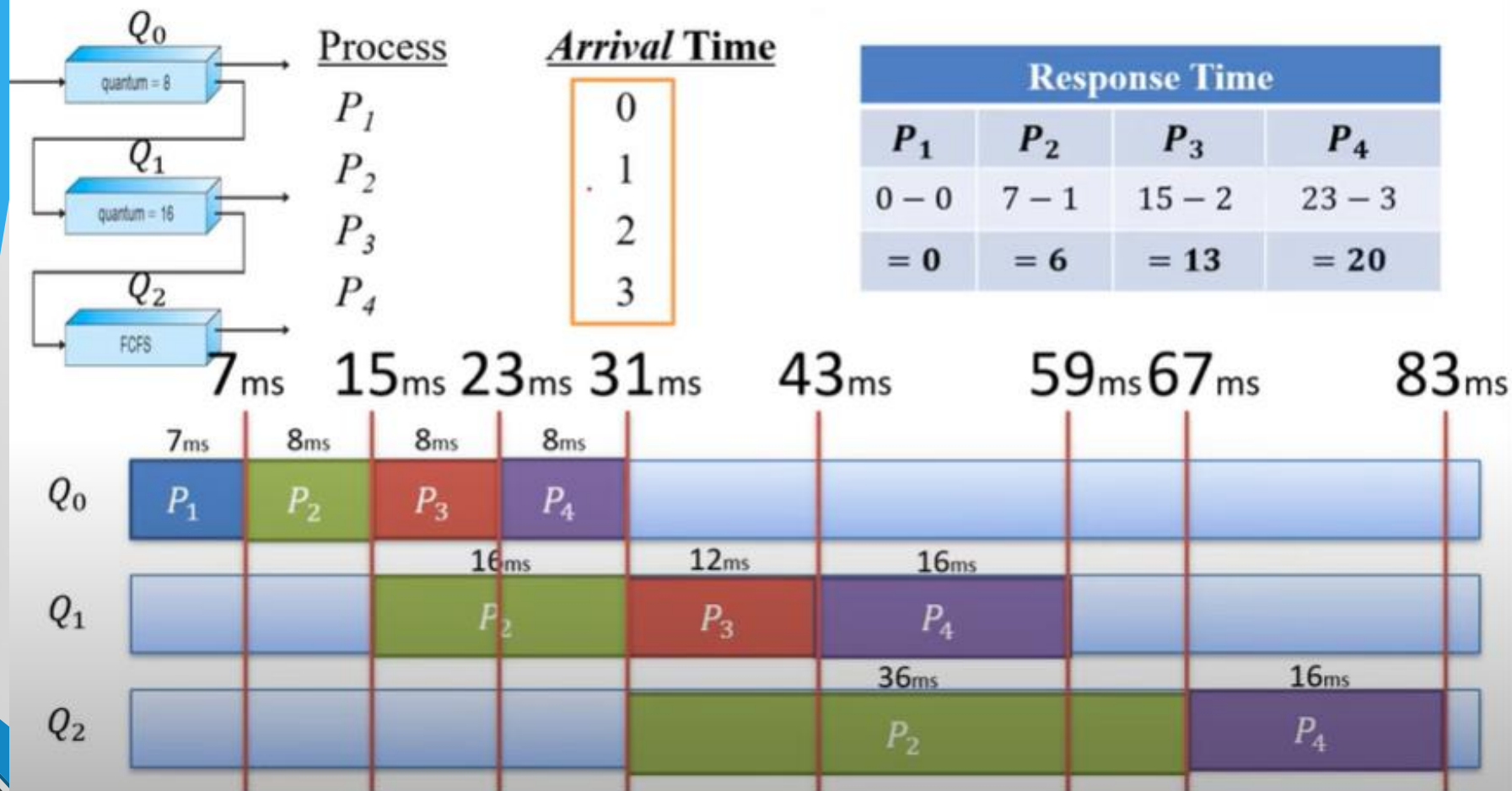
Example of Multilevel Feedback Queue



Example of Multilevel Feedback Queue



Example of Multilevel Feedback Queue





Thank You